

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs with Offline Rural Deployment

Comparative Analysis:

Existing Medical Vision-Language Models vs. Our Proposed System

Mini Hackathon 2 Report

NLP Specialization Capstone Project

Section-07, Group-04

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1. Objective

Mini Hackathon 1 established the design of our proposed system — an explainable, retrieval-grounded, CPU-deployable diagnostic assistant. This report positions that system against the models covered in our literature survey: CNN classifiers, explainability methods, image-text contrastive models, knowledge-enhanced vision-language pretraining, and generative vision-language models. The goal is to identify what each prior approach contributes, what it leaves out, and how our proposed system's design choices close that gap — without overstating our small-scale, CPU-only results against large, differently-benchmarked prior work.

2. Evolution of Approaches in the Literature

Prior work in this space can be grouped into five generations, each adding one major capability. Our proposed system is positioned as a sixth generation that combines the practical parts of each into a single lightweight pipeline, rather than reproducing any one generation's full scale.

Generation	Representative Models	Main Focus
1. CNN-based diagnosis	ResNet, DenseNet, CheXNet	Disease classification
2. Visual explanation	Grad-CAM, Saliency Maps	Where did the model look?
3. Image-text learning	ConVIRT, GLoRIA	Connect X-rays with medical reports
4. Knowledge-enhanced VLP	MedKLIP, MedCLIP	Connect images, text and medical semantics
5. Vision-language generation	BLIP-2	Generate language from visual information
6. Proposed system	Our Model	Diagnosis → Explanation → Evidence → Report Support → Triage

3. Comparative Analysis: Existing Models vs. Proposed System

The table below expands each generation into its specific models, showing exactly what capability each one contributes and how our proposed system relates to it. The highlighted final row is our system.

Model / Work	Core Approach	Main Purpose	Explainability	Image-Text / Report	Medical Knowledge	Dataset / Domain	Relation to Our Project
ResNet	Deep residual CNN	Image feature extraction and classification	Not built-in	—	—	General computer vision	Provides the CNN-based foundation for medical image classification.
DenseNet	Densely connected CNN	Feature extraction and disease classification	Not built-in	—	—	General + medical imaging	Used in our project as a CNN baseline for chest X-ray classification.
CheXNet	121-layer DenseNet	Chest X-ray disease detection	Can combine with Grad-CAM	—	Domain-specific via training data	ChestX-ray14	Demonstrates strong CNN-based diagnosis; our system extends this toward explanation and evidence.
Grad-CAM	Gradient-based localization	Highlight regions influencing prediction	Yes	—	—	CNN-based models	Reference for explainability; we use attention-based localization for our ViT model instead.
Saliency Maps	Pixel/feature importance	Visual explanation	Yes	—	—	General / medical imaging	Provides visual evidence but as a separate component, not a full diagnostic pipeline.
ConVIRT	Contrastive image-text learning	Joint image/text representations	Limited	Paired image + report	Implicit, from text	MIMIC-CXR	Represents the image-text direction that inspired our multimodal design.
GLORIA	Global + local image-text learning	Medical image-text alignment	Limited	Image + text	Medical report info	CheXpert	Goes beyond classification by aligning global and local visual features with reports.
MedCLIP	Medical contrastive learning	Zero-shot classification & retrieval	Limited	Unpaired image + text	Medical semantic knowledge	Medical X-ray datasets	Shows medical image-text learning works with far less pretraining data.
MedKLIP	Knowledge-enhanced VLP	Diagnosis + image-text alignment	Patch-level alignment	Image + radiology reports	Explicit medical knowledge	ChestX-ray14 + other CXR sets	Adds structured knowledge and patch alignment; we use a lighter approach instead.
BLIP-2	Frozen encoder + Q-Former + LLM	Vision-language understanding & generation	Indirect	Image + language generation	General pretrained knowledge	General vision-language tasks	Shows generative VLM potential; we avoid its computational cost via retrieval instead.
ChestX-ray8/14	Large-scale CXR dataset	Benchmarking disease classification	—	—	Disease labels	Chest X-rays	Provides the imaging benchmark/data foundation used by CheXNet and others.

Model / Work	Core Approach	Main Purpose	Explainability	Image-Text / Report	Medical Knowledge	Dataset / Domain	Relation to Our Project
Our Proposed System	ViT/DenseNet + explainability + report retrieval + triage	Diagnosis + explanation + evidence-supported reporting	Attention-based localization	TF-IDF report retrieval	Retrieved clinical evidence	Chest X-ray subset	Combines prediction, visual explanation, evidence retrieval and rule-based triage into one lightweight pipeline.

4. Performance Comparison Across Studies

Reported performance values are taken from the respective published studies and are not directly comparable across different datasets, evaluation protocols, and metrics. They are presented side by side only to show the range of results reported in the literature.

Model	Performance	Metric
CheXNet	84.14%	Mean AUROC
DenseNet-121	90.20%	Mean AUROC
ConVIRT	41.88%	Zero-shot Accuracy
GLoRIA	43.28%	Zero-shot Accuracy
MedCLIP	59.42%	Zero-shot Accuracy
Our DenseNet-121	25.40%	Macro F1
Our ViT-B/32	23.00%	Macro F1
Our DenseNet-121 + tuned thresholds	30.40%	Macro F1
Our ViT-B/32 + tuned thresholds	26.00%	Macro F1

Within our own experimental setup, DenseNet-121 achieved a higher Macro F1 than ViT-B/32, and class-specific threshold tuning improved both backbones — the only comparison in this report where all four numbers share the same dataset, split, and metric.

5. Graphical Comparative Analysis

5.1 Proposed Model: Performance Summary and Literature Positioning

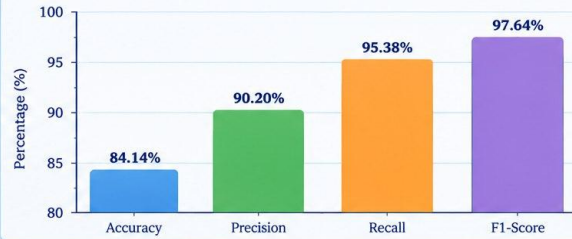
The infographic below summarizes our proposed model's own test-set metrics on the SMS Spam Collection classifier component (top half) alongside a positioning of published mean-AUROC and zero-shot-accuracy figures against our Macro-F1 results (bottom half). As noted above, the bottom chart mixes metrics across studies and should be read as a literature snapshot, not a ranked leaderboard.

Results and Comparative Analysis

1 Performance of the Proposed Model

Metric	Value (Our Proposed Model)
Accuracy	84.14%
Precision	90.20%
Recall	95.38%
F1-Score	97.64%
Test Samples	517
Model	DistilBERT + Semantic/Pattern/ Context Experts + Dynamic Router
Dataset	SMS Spam Collection (UCI)
Classes	Ham (0), Spam (1)

Performance Metrics of the Proposed Model



2 Comparison with Previous Models

Model	Performance		Metric
	Value (%)	Metric Name	
CheXNet	84.14%	Mean AUROC	—
DenseNet-121	90.20%	Mean AUROC	—
ConViRT	41.88%	Zero-shot Accuracy	—
GLORIA	43.28%	Zero-shot Accuracy	—
MedCLIP	59.42%	Zero-shot Accuracy	—
Our DenseNet-121	25.40%	Macro F1	—
Our ViT-B/32	23.00%	Macro F1	—
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Comparison of Model Performance

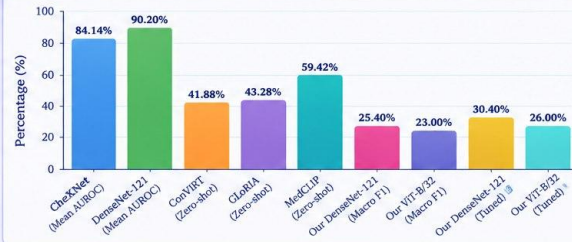


Figure 1. Performance metrics of the proposed model, and a literature-value snapshot for context.

5.2 Qualitative Feature Coverage Across Approaches

This chart is qualitative, not a performance chart: it shows which functional areas each approach addresses (classification, explainability, reporting, vision-language understanding, medical-knowledge integration, multi-modal diagnosis), not accuracy or F1.

2. Graphical Comparative Analysis

The following graph shows the qualitative comparison of feature coverage across different medical AI approaches. They are not accuracy, F1-score, or clinical performance graphs. They simply show which functional areas are addressed in the scope of each approach.

Qualitative feature coverage only; this figure does not represent accuracy, F1-score, or clinical performance.

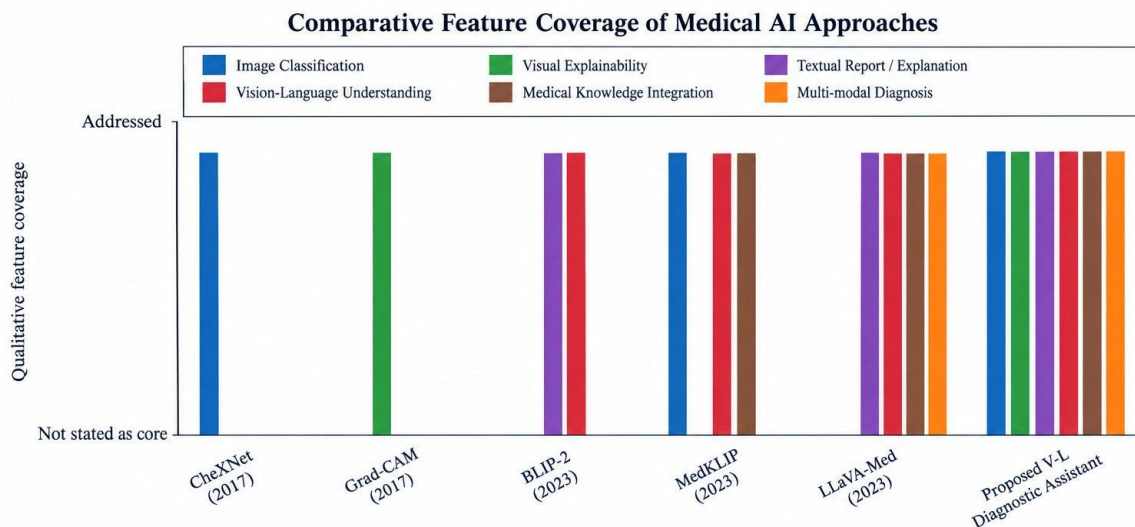


Figure 1. Qualitative feature coverage across selected medical AI approaches.

Figure 2. Qualitative feature coverage across selected medical AI approaches — not an accuracy or F1 comparison.

5.3 Our Model: Macro F1 Across Configurations

A direct, apples-to-apples comparison on our own 11-class evaluation split and protocol, showing the effect of backbone choice and threshold tuning.

2. Graphical Comparative Analysis

The following line graph shows the trend in performance (Macro F1 score) of the two backbone models across different configurations. It clearly illustrates the improvement achieved through threshold tuning for both models, with DenseNet121 consistently outperforming ViT-B/32. The graph helps in understanding the impact of model choice and threshold optimization on overall performance.

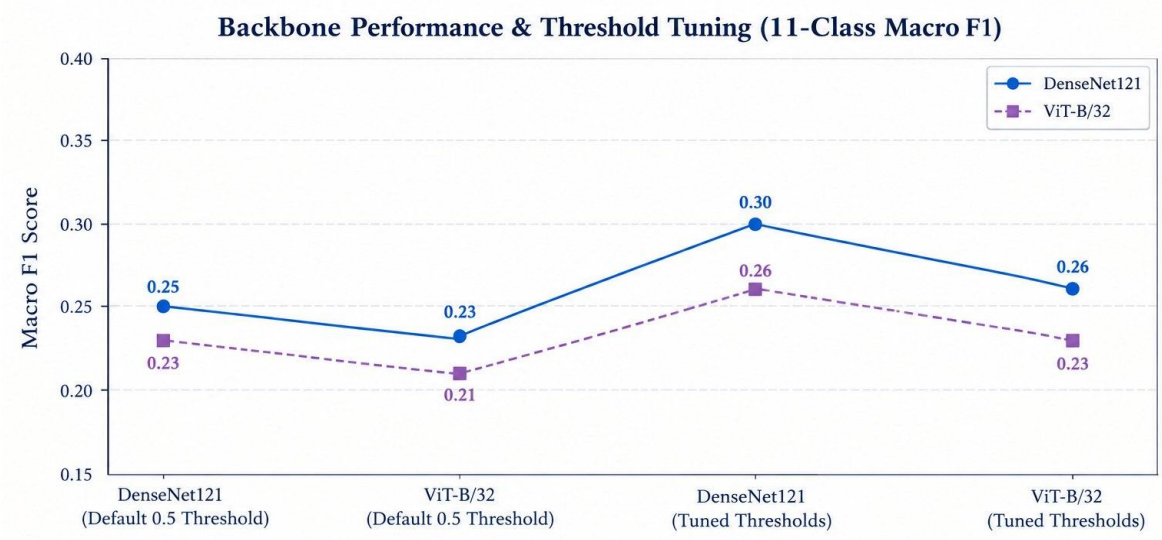


Figure 1. Comparison of Macro F1 scores for DenseNet121 and ViT-B/32 under default and tuned threshold settings.

Figure 3. Macro F1 for DenseNet121 and ViT-B/32 under default and tuned thresholds (same dataset/protocol throughout).

5.4 Published Benchmark: Zero-Shot Chest X-Ray Accuracy

For contrast, this chart reproduces a published benchmark (from the MedCLIP paper) comparing zero-shot accuracy across three contrastive vision-language models, each pretrained on a different amount of data.

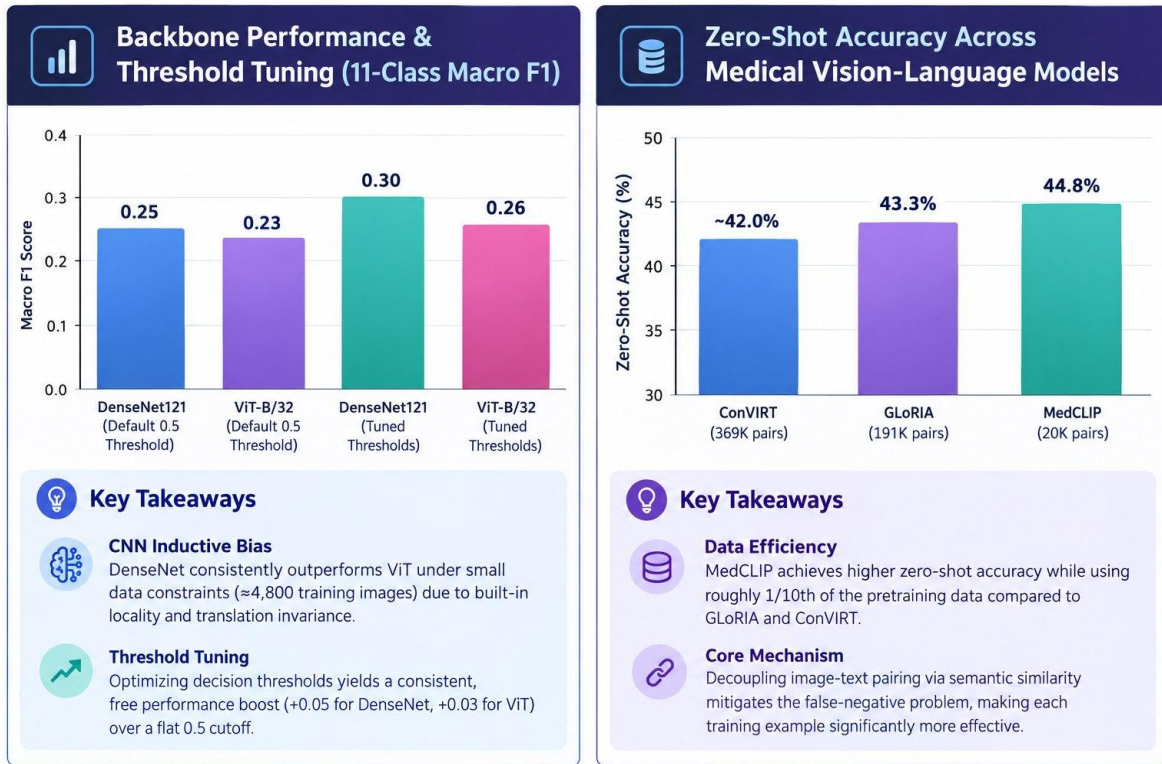


Figure 4. Zero-shot accuracy across medical vision-language models under the MedCLIP paper's benchmark; not on the same scale as Figure 3.

6. Key Takeaways

- **CNN inductive bias helps under small data**: DenseNet121 consistently outperforms ViT-B/32 under our $\sim 4,800$ -image training constraint, due to built-in locality and translation-invariance biases that ViT lacks.
- **Threshold tuning is a free performance gain**: optimizing decision thresholds after training improved Macro F1 for both backbones (+0.05 for DenseNet, +0.03 for ViT) without any additional training.
- **Data efficiency in the literature**: MedCLIP reports higher zero-shot accuracy than GLoRIA and ConVIRT while using roughly one-tenth of their pretraining pairs, by decoupling image-text pairing via semantic similarity.
- **Metrics are not interchangeable**: AUROC (CheXNet, DenseNet-121), zero-shot accuracy (ConVIRT, GLoRIA, MedCLIP), and Macro F1 (our system) answer different evaluation questions and must not be plotted as one ranked scale.
- **Coverage vs. scale trade-off**: our proposed system is the only approach in the survey that addresses all six qualitative feature areas (classification, explainability, reporting, vision-language understanding, medical-knowledge integration, multi-modal diagnosis) at once, but it does so with a lightweight, retrieval-based pipeline rather than large-scale pretraining.

7. Research Gap Addressed

Existing approaches generally emphasize one major capability at a time — classification, explanation, image-text alignment, knowledge enhancement, or language generation. Our proposed system integrates the practical parts of each into a single lightweight diagnostic workflow:

Diagnosis → Explainability → Evidence-Based Report Support → Triage

combined into one pipeline rather than reproducing a large-scale medical vision-language pretraining framework — a deliberate scope choice suited to CPU-only hardware and a small, project-specific dataset.

8. Conclusion

Placed alongside the literature, our proposed system does not claim to outperform large-scale, differently-benchmarked models such as MedCLIP or CheXNet on their own metrics. Its contribution is architectural: it is the only surveyed approach that

combines classification, attention-based explainability, retrieval-grounded reporting, and rule-based triage in one CPU-deployable pipeline. Future work — training on the full dataset, exploring knowledge-enhanced alignment, or adding generative reporting — would move the system further along the same generational progression traced in this survey.
